

Full support of the zero-temperature Sherrington–Kirkpatrick order parameter

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Abstract

We prove that every admissible integrable minimizer of the zero-temperature Parisi functional for the pure, zero-field Sherrington–Kirkpatrick model has full relative Stieltjes support on $[0, 1]$. Thus its support has no gaps at any overlap scale below one. We use the covariance normalization $\xi(t) = t^2/2$.

1 Introduction

For independent standard Gaussian variables J_{ij} and spins $\sigma \in \{-1, 1\}^n$, the pure Sherrington–Kirkpatrick (SK) Hamiltonian with no external field is

$$H_n(\sigma) = \frac{1}{\sqrt{n}} \sum_{i < j} J_{ij} \sigma_i \sigma_j.$$

Its ground-state problem asks for the limiting maximum energy per spin. The Parisi variational formula expresses this value through an order parameter encoding a hierarchy of overlap scales. Infinitely many scales need not fill an interval; the question studied here is whether every scale below one belongs to the zero-temperature support.

We first specify the variational problem, including the convention at the left endpoint. Let $\mathcal{U}$ be the set of nonnegative, nondecreasing, right-continuous functions $\gamma : [0, 1] \rightarrow [0, \infty)$ satisfying $\int_0^1 \gamma(t) dt < \infty$. The Stieltjes measure μ_γ on $[0, 1]$ is defined by

$$\mu_\gamma([0, t]) = \gamma(t), \quad 0 \leq t < 1.$$

In particular $\mu_\gamma(\{0\}) = \gamma(0)$. It is locally finite, but need not be a probability measure or have finite total mass. All support statements below use the relative topology of $[0, 1]$.

For $\gamma \in \mathcal{U}$, define the Parisi value function by

$$\Phi_\gamma(t, x) = \sup_{|a| \leq 1} \mathbb{E} \left[\left| x + B_1 - B_t + \int_t^1 \gamma(s) a_s ds \right|^2 - \frac{1}{2} \int_t^1 \gamma(s) a_s^2 ds \right]. \quad (1.1)$$

Here B is standard Brownian motion and the supremum is over progressively measurable controls $a_s \in [-1, 1]$, adapted to its increments after time t . This is the solution, obtained by monotone-coefficient approximation, of

$$\Phi_t = -\frac{1}{2}(\Phi_{xx} + \gamma(t)\Phi_x^2), \quad \Phi(1, x) = |x|. \quad (1.2)$$

Section 2 proves the regularity and approximation facts we use, directly from the heat equation and (1.1). The functional to be minimized is

$$\mathcal{P}(\gamma) = \Phi_\gamma(0, 0) - \frac{1}{2} \int_0^1 t \gamma(t) dt. \quad (1.3)$$

This normalization has covariance $\xi(t) = t^2/2$, hence $\xi'' = 1$. For two spin configurations with overlap $R_{12} = n^{-1} \sum_i \sigma_i^1 \sigma_i^2$, the covariance is $\mathbb{E} H_n(\sigma^1) H_n(\sigma^2) = n R_{12}^2/2 - 1/2$. Adding an independent centered Gaussian of variance $1/2$, constant across spin configurations, gives exactly $n \xi(R_{12})$ and leaves the expected maximum unchanged. The zero-temperature Parisi formula and its attainment give¹

$$P_* := \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E} \max_{\sigma} H_n(\sigma) = \min_{\gamma \in \mathcal{U}} \mathcal{P}(\gamma). \quad (1.4)$$

We use this established formula and existence theorem from Auffinger and Chen [3, Theorem 1 and the proof of Lemma 3]. The proof of the support assertion itself applies to any minimizer of the explicitly defined functional (1.3).

Write $u_\gamma = \Phi_{\gamma, x}$ and define its controlled diffusion by

$$dX_t = \gamma(t) u_\gamma(t, X_t) dt + dB_t, \quad X_0 = 0. \quad (1.5)$$

The drift is bounded and Lipschitz in space on every compact time strip; its absolute value is bounded by the integrable function γ . Thus this equation has a unique strong solution up to time one, as proved in Lemma 2.2.

Theorem 1.1 (Full support at zero temperature). *If $\gamma \in \mathcal{U}$ minimizes $\mathcal{P}$, then*

$$\text{supp } \mu_\gamma = [0, 1].$$

In particular $\gamma(b) > \gamma(a)$ whenever $0 \leq a < b < 1$. For the solution of (1.5),

$$\mathbb{E}[u_\gamma(t, X_t)^2] = t, \quad \mathbb{E}[\Phi_{\gamma, xx}(t, X_t)^2] = 1, \quad 0 \leq t < 1. \quad (1.6)$$

Chen [7, Theorem 1.3] proves the same full-relative-support conclusion, together with smooth absolute continuity of the minimizing Stieltjes measure. The present proof excludes gaps through an explicit rational coercive identity and an unconditional strict-curvature inequality on positive constant intervals. Corollary 6.4 recovers $\gamma \in C^\infty([0, 1])$ and $\gamma(0) = 0$ by the consistency and polynomial-moment argument of [7, Proposition 4.6 and Appendix B.6]. A fixed-time approximation by constant intervals then transfers our coercive bound to the minimizer and gives $\gamma'(t) > 0$ for $0 < t < 1$. Thus $\mu_\gamma = \gamma'(t) dt$ has a smooth density which is strictly positive away from zero. We also develop the scalar martingale value representation and finite Gaussian coefficients in Section 7, and apply the optimization theorem of El Alaoui, Montanari, and Sellke in Corollary 7.3.

Zero external field is used in the evenness of Φ and the initial condition $X_0 = 0$; the constant value of ξ'' is used in the curvature identities. Neither a nonzero-field extension nor an extension to a general mixed-spin covariance is implicit in the statement.

¹The normalized maximum is a Lipschitz function of the Gaussian disorder with constant $\sqrt{(n-1)/(2n^2)}$. Gaussian concentration makes its difference from its expectation tend to zero in probability. Together with the almost-sure ground-state limit in the cited formula, this implies convergence of the expectations to the same constant.

1.1 Prior work and the distinction between support assertions

Sherrington and Kirkpatrick introduced the infinite-range Gaussian Ising spin-glass model [17]. Parisi proposed a hierarchy of replica-symmetry breaking described by a functional order parameter [15]; its interpretation through overlaps was developed subsequently [16]. Guerra established the interpolation bound for the Parisi variational expression [10], and Talagrand proved the Parisi formula for the SK free energy at every positive temperature [18]. Identifying the support of the minimizing order parameter is a separate question from proving that variational formula.

The zero-temperature variational problem is part of the rigorous Parisi theory developed by Auffinger and Chen [3]. The control formulation has antecedents in the Brownian variational representation of Boué and Dupuis [5] and the diffusion-control representation of the Parisi equation in Bovier and Klimovsky [6, Section 6.2]. Auffinger and Chen [2] and Jagannath and Tobasco [11] developed stochastic-control, strict-convexity, and gradient-martingale arguments for this equation. Chen, Handschy, and Lerman [8, Propositions 2–4, Theorems 4–5, and Lemma 3] give zero-temperature regularity, control, uniqueness, and variation interfaces. We prove the analytic and variational facts needed here, including global-in-space derivative convergence on compact time strips, rather than presume that a smooth-terminal statement applies at the terminal cusp $|x|$.

Auffinger, Chen, and Zeng proved that the zero-temperature SK order parameter has infinitely many support points [4, Theorem 1]. They also recorded the exclusion of a terminal constant interval [4, Remark 7]; Section 6 supplies a detailed boundary-layer proof of that endpoint fact. Infinite support and absence of a terminal plateau do not exclude an interior gap. The main issue here is the elimination of every such gap.

At positive temperature, Zhou [19, Theorem 1] established an interval-support result near the critical temperature, and Lopatto [13, Theorem 1.1] established full interval support throughout the zero-field SK low-temperature phase. Lopatto’s proof propagates inequalities for Cole–Hopf profiles and a transformed diffusion density before applying a constant-interval crossing argument [13, Sections 3, 7–9]. Chen’s zero-temperature analysis also uses transformed-density inequalities and smooth-terminal approximation [7, Propositions 3.7 and 3.11]. The crossing conclusions in [13, Proposition 9.1] and [7, Proposition 4.2] are conditional: if the second derivative of the gradient’s second moment vanishes at a point of a positive constant interval, its third derivative is positive at that point. Our rational identity instead bounds that third derivative below by a strictly positive quantity throughout each such interval. The argument is carried out directly at zero temperature; interval support alone does not in general pass through weak limits of measures.

The variational contact conditions have precedents in both the positive-temperature analysis of Jagannath and Tobasco [12, Proposition 1.1, Lemma 3.2, and Corollaries 3.6 and 3.10] and the zero-temperature analysis of Chen, Handschy, and Lerman [8, Propositions 3–4]. In the present setting one may add or remove Stieltjes mass without a probability normalization constraint. This gives a contact potential which is nonnegative and vanishes on the support, a fact we derive explicitly. Once consistency is saturated at every time, differentiating polynomial moments yields regularity of the order parameter. This method appears in Auffinger and Chen [1, Lemma 2 and the proof of Theorem 2(ii)] and in the zero-temperature form of Chen [7, Appendix B.6].

The scalar consequences in Section 7 follow the incremental approximate-message-passing methodology of Montanari [14] and El Alaoui, Montanari, and Sellke [9]: a Hessian-weighted gradient martingale determines normalized Euler increments and the limiting value. We prove the needed scalar statements directly. The finite-disorder algorithm is supplied separately by the theorem of El Alaoui, Montanari, and Sellke [9, Theorem 3 and Corollary 2.2 of the cited preprint]: full support verifies its strictly increasing minimizer hypothesis. Corollary 7.3 spells out the Gaussian input normalization and the resulting $C(\epsilon)n^2$ real-arithmetic operation count.

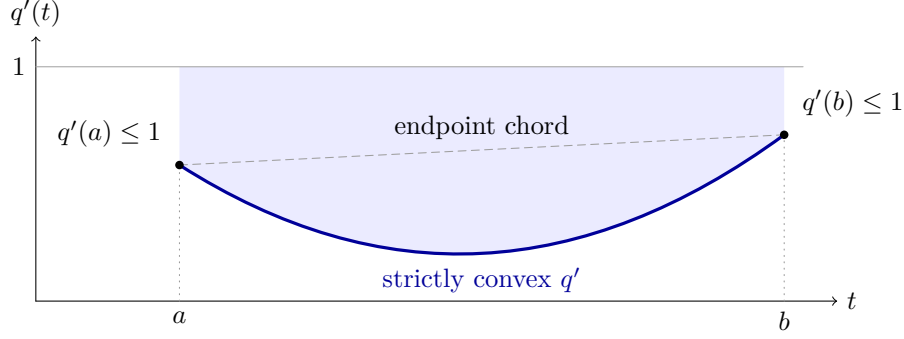


Figure 1: The contradiction on a hypothetical support gap with $\gamma = c > 0$. Strict convexity puts q' below its endpoint chord, which is at most 1. The shaded deficit is positive, whereas contact at both endpoints requires $\int_a^b q'(t) dt = q(b) - q(a) = b - a$. The endpoint derivatives need not equal 1.

1.2 Proof structure

The argument has an analytic part and a variational part. Section 2 first constructs the diffusion and proves the compact-strip estimates needed to approximate integrable order parameters. Set $q(t) = \mathbb{E}u_\gamma(t, X_t)^2$. The gradient-martingale identity gives $q'(t) = \mathbb{E}\Phi_{\gamma,xx}(t, X_t)^2$. At a minimizer, the first variation shows that

$$G(s) = \int_s^1 (q(t) - t) dt$$

is nonnegative and vanishes on $\text{supp } \mu_\gamma$. At an interior support point this implies $q(t) = t$ and $q'(t) \leq 1$. On a hypothetical interior gap (a, b) with $\gamma = c > 0$, the endpoint conditions give $q(a) = a$, $q(b) = b$, and $q'(a), q'(b) \leq 1$. If q' is strictly convex there, it lies below its endpoint chord and hence below 1 inside. Integrating gives $q(b) - q(a) < b - a$, contradicting contact at the endpoints, as illustrated in Figure 1.

The main work is to establish this strict convexity without assuming support properties. For a positive constant coefficient c , put $r = c\Phi_x$, $v = r_x$, $w = r_{xx}$, and $z = r_{xxx}$. Direct differentiation along the diffusion yields

$$q'''(t) = c^{-2} \mathbb{E}[z^2 - 12vw^2 + 6v^4](t, X_t).$$

The expression inside this expectation has a negative term and is not controlled pointwise by convexity of Φ alone. Sections 3 and 4 overcome this obstacle. For positive finite-step coefficients with smooth terminal data, let $f(t, \cdot)$ be the density of X_t . The factorization $f = e^{c\Phi}g$ gives a forward heat equation for g between jumps. We propagate backward shape inequalities for r and forward inequalities for the score $s = -\partial_x \log g$, including a comparison of s/r . These signs make an exact rational polynomial identity nonnegative after one spatial integration by parts, retaining the lower bound $10^{-3}\mathbb{E}v^4$.

Section 5 transfers only this integrated inequality to an arbitrary integrable coefficient; it does not require convergence of the auxiliary density scores or their ratios. The zero-coefficient case follows separately from the heat equation. Section 6 then excludes interior, initial, and terminal gaps. With $q(t) = t$ at every time, the Hessian Itô identity expresses γ as a quotient of continuous derivative moments. Differentiating polynomial moments simultaneously bootstraps this formula to smoothness. To obtain $\gamma'(t) > 0$ at a fixed $t > 0$, we approximate γ by coefficients which equal $\gamma(t)$ on shrinking intervals around t . The integrated coercive inequality passes to the limit at that fixed time and becomes a strictly positive lower bound for $\gamma'(t)$. Finally, Section 7 derives a martingale

formula for P_* and a finite Gaussian approximation, then verifies the hypothesis and normalization of the existing optimization theorem.

2 Integrable coefficients and the controlled diffusion

The sign inequalities will first be proved for finite step functions. We therefore need estimates which survive approximation of an arbitrary $\gamma \in \mathcal{U}$, including coefficients which diverge at time one. All estimates in this section are on strips ending strictly before one. The control and heat-semigroup methods follow the Parisi PDE framework of Auffinger and Chen [2], Jagannath and Tobasco [11], and the zero-temperature treatment of Chen, Handschy, and Lerman [8, Proposition 2 and Theorem 5]. We include the proofs to make the approximation uniformity explicit. We retain the notation $u_\gamma = \partial_x \Phi_\gamma$ from the model definition, and write P_t for the heat semigroup with variance t .

Lemma 2.1 (Compact-strip regularity and stability). *For every $\gamma \in \mathcal{U}$, the function Φ_γ is even, convex, and one-Lipschitz in its spatial variable. It satisfies*

$$|x| \leq \Phi_\gamma(t, x) \leq |x| + \sqrt{\frac{2(1-t)}{\pi}} + \frac{1}{2} \int_t^1 \gamma(s) ds. \quad (2.1)$$

For each integer $k \geq 1$ and $T < 1$, the derivative $\partial_x^k \Phi_\gamma$ is bounded and jointly continuous on $[0, T] \times \mathbb{R}$. These bounds are uniform when the coefficients have uniformly bounded $L^1(0, 1)$ norms. The derivatives are locally absolutely continuous in time and satisfy the spatially differentiated Parisi equation for almost every time.

If $\gamma_n \rightarrow \gamma$ in $L^1(0, 1)$, then, for every $k \geq 1$ and $T < 1$,

$$\sup_{[0, T] \times \mathbb{R}} |\partial_x^k \Phi_{\gamma_n} - \partial_x^k \Phi_\gamma| \longrightarrow 0. \quad (2.2)$$

The same conclusion holds when each γ_n is a finite step function and its terminal datum is $M_n^{-1} \log(2 \cosh(M_n x))$, where $M_n \rightarrow \infty$.

Proof. For a nonnegative finite step coefficient and a smooth even convex one-Lipschitz terminal datum ψ , backward evolution across an interval with coefficient c is

$$\Phi(t, x) = \begin{cases} c^{-1} \log P_{b-t}(e^{c\Phi(b, \cdot)})(x), & c > 0, \\ P_{b-t}\Phi(b, \cdot)(x), & c = 0. \end{cases} \quad (2.3)$$

These formulas preserve evenness, convexity, and the one-Lipschitz bound. Convexity in the first case follows either from Hölder's inequality applied after a spatial translation, or by differentiating the logarithm: its second derivative is the tilted mean of the old second derivative plus c times the tilted variance of the old first derivative. The first derivative is a tilted mean of the old first derivative. The assertions in the case $c = 0$ follow directly from convolution.

For these step solutions, Itô's formula and completion of the square give the control representation

$$\Phi(t, x) = \sup_{|a| \leq 1} \mathbb{E} \left[\psi \left(x + B_1 - B_t + \int_t^1 \gamma(s) a_s ds \right) - \frac{1}{2} \int_t^1 \gamma(s) a_s^2 ds \right]. \quad (2.4)$$

The supremum is over progressively measurable controls. Indeed, the drift of $\Phi(s, Y_s) - \frac{1}{2} \int_t^s \gamma a^2$ is $-\gamma(a - u_\gamma)^2/2$ when $dY_s = \gamma(s) a_s ds + dB_s$. Equality is attained by $a_s = u_\gamma(s, Y_s)$; for the smooth

finite-step problem the feedback equation has a unique strong solution. Consequently two such solutions obey

$$\|\Phi_{\gamma,\psi} - \Phi_{\tilde{\gamma},\tilde{\psi}}\|_{\infty} \leq \|\psi - \tilde{\psi}\|_{\infty} + \frac{3}{2}\|\gamma - \tilde{\gamma}\|_1. \quad (2.5)$$

Here the supremum is over the whole time-space domain, even though the individual functions grow linearly. The bound follows by using the same controls on both sides of (2.4).

Nonnegative nondecreasing finite step functions are dense in $\mathcal{U}$ for the L^1 norm. For example, truncate at a time increasing to one, approximate the resulting bounded monotone function by monotone step functions, and then let the truncation time tend to one. Since $0 \leq M^{-1} \log(2 \cosh(Mx)) - |x| \leq M^{-1} \log 2$, (2.5) defines a unique uniform limit, with control representation (2.4) and terminal datum $|x|$. This agrees with the Parisi solution in the model definition. All the asserted shape properties pass to the limit. The zero control and Jensen's inequality give the lower bound in (2.1). For the upper bound use the triangle inequality and $|a| - a^2/2 \leq 1/2$ in the control representation.

We give the smoothing estimates explicitly. This also establishes the uniformity needed below. On any strip $[0, b]$ with $b < 1$, monotonicity gives

$$\gamma(t) \leq \frac{\|\gamma\|_1}{1-b}, \quad 0 \leq t \leq b. \quad (2.6)$$

Thus all coefficients in an L^1 -bounded family have a common bound Γ there. Work first with the smooth-terminal step solutions and put $U = \Phi_x$. Differentiation of the mild equation gives

$$U(t) = P_{b-t}U(b) + \int_t^b \gamma(s)P_{s-t}[U(s)U_x(s)] ds. \quad (2.7)$$

We always have $\|U\|_{\infty} \leq 1$, independently of its terminal smoothing. The heat-kernel estimate $\|\partial_x P_h f\|_{\infty} \leq Ch^{-1/2}\|f\|_{\infty}$ implies

$$\|U_x(t)\|_{\infty} \leq \frac{C}{\sqrt{b-t}} + C\Gamma \int_t^b \frac{\|U_x(s)\|_{\infty}}{\sqrt{s-t}} ds. \quad (2.8)$$

The constants do not involve derivatives of the terminal datum.

For completeness, the integral inequality in (2.8) gives a uniform bound away from b . The convolution of $h^{-1/2}$ with itself equals π . Substituting the inequality once into its integral therefore reduces it to an ordinary backward Gronwall inequality, with forcing bounded by a constant times $1 + (b-t)^{-1/2}$. That forcing is integrable. The resulting estimate is uniform on every smaller strip.

The same argument bounds all higher derivatives. Suppose that derivatives of U through order $m-1$ are already bounded on a slightly larger strip. Differentiate (2.7) m times, placing one derivative on the heat kernel and the other $m-1$ on UU_x . Its highest-order term is $U\partial_x^m U$; every remaining product involves derivatives of orders at most $m-1$. The initial term is bounded by $C(b-t)^{-1/2}\|\partial_x^{m-1}U(b)\|_{\infty}$. Thus the same integral inequality applies to $\|\partial_x^m U(t)\|_{\infty}$, with an additional bounded forcing. Choose successively smaller strip endpoints in this induction. For every fixed m this proves the asserted bound, uniformly for the approximants.

Uniform convergence of the functions now gives convergence of all their spatial derivatives without a separate singular-kernel comparison. If a bounded C^2 function F on $\mathbb{R}$ has bounded second derivative, Taylor's formula gives, for every $h > 0$,

$$\|F'\|_{\infty} \leq \frac{2\|F\|_{\infty}}{h} + \frac{h}{2}\|F''\|_{\infty}. \quad (2.9)$$

Apply this to a difference of two approximating solutions, uniformly on a fixed smaller strip. The uniform second-derivative bound and uniform convergence imply convergence of the first derivatives.

Apply (2.9) successively to the differences of the first, second, and later derivatives; the next two derivative bounds are available at every stage. This proves uniform convergence of every positive spatial derivative. Their limits are the spatial derivatives of the limiting function, by the fundamental theorem of calculus. Since the approximating derivatives are jointly continuous, so are their uniform limits.

The argument applies also to an arbitrary L^1 -convergent sequence in $\mathcal{U}$: first obtain these bounds for each member by step approximation, then use (2.5), the uniform local coefficient bound (2.6), and (2.9). Finally pass to the limit in the integral form of the differentiated equation on each compact strip. All its spatial factors converge uniformly, and its coefficient converges in L^1 . This proves local absolute continuity in time and the asserted almost-everywhere equations. $\square$

Lemma 2.2 (Diffusion and square-martingale identity). *For $\gamma \in \mathcal{U}$, the equation*

$$dX_t = \gamma(t)u_\gamma(t, X_t) dt + dB_t, \quad X_0 = 0, \quad (2.10)$$

has a unique strong solution on $[0, 1]$. On every strip $[0, T]$ with $T < 1$,

$$u_\gamma(t, X_t) = \int_0^t \Phi_{\gamma,xx}(s, X_s) dB_s. \quad (2.11)$$

In particular, $q(t) = \mathbb{E}[u_\gamma(t, X_t)^2]$ belongs to $C^1[0, 1]$, $q(0) = 0$, and

$$q'(t) = \mathbb{E}[\Phi_{\gamma,xx}(t, X_t)^2]. \quad (2.12)$$

Under the approximations in Lemma 2.1, the coupled diffusions converge uniformly on compact time strips, and q_n and q'_n converge uniformly there.

Proof. The drift is measurable in time, bounded by $\gamma(t)$, and globally Lipschitz in space on every compact time strip by Lemma 2.1. Picard iteration gives a unique strong solution there. These solutions agree on overlaps and extend to time one because $\int_0^1 \gamma < \infty$ and Brownian paths are continuous.

For clarity, convergence needs no moment estimate for the approximating paths. Couple them using the same Brownian motion. The drift difference is bounded by a common spatial Lipschitz constant times $|X_n - X|$, plus $|\gamma_n - \gamma| + \gamma_n \|u_n - u_\gamma\|_\infty$. Gronwall's inequality gives a deterministic bound for $\sup_{t \leq T} |X_n(t) - X(t)|$ tending to zero. For step coefficients and smooth terminal data, Itô's formula on each step cancels the drift of $u_n(t, X_n(t))$. The step-boundary values agree. Passing to the limit in the stochastic integral by its L^2 isometry proves (2.11). Its initial value is zero by evenness. The same isometry gives $q(t) = \int_0^t \mathbb{E}[\Phi_{\gamma,xx}(s, X_s)^2] ds$. The integrand is continuous by joint continuity, boundedness on compact strips, and continuity of X . This proves (2.12) and the stated convergence assertions. $\square$

Lemma 2.3 (Transfer on a constant interval). *Suppose $\gamma = c$ on an open interval $I \subset (0, 1)$. If $c > 0$, every point of I has a neighborhood on which one may choose positive, bounded, nondecreasing finite step approximants equal to c , with smooth terminal data and $M_n \geq \|\gamma_n\|_\infty$. At each time in this neighborhood, all spatial derivatives and the corresponding diffusion expectations of bounded polynomials in those derivatives converge. Moreover, $\Phi_{\gamma,xx} > 0$ throughout $I \times \mathbb{R}$.*

For $c > 0$, set $r = c\Phi_x$, $v = r_x$, $w = r_{xx}$, and $z = r_{xxx}$. Then, within I ,

$$(q')'(t) = c^{-2} \mathbb{E}[w(t, X_t)^2 - 2v(t, X_t)^3], \quad (2.13)$$

$$(q'')'(t) = c^{-2} \mathbb{E}[z(t, X_t)^2 - 12v(t, X_t)w(t, X_t)^2 + 6v(t, X_t)^4]. \quad (2.14)$$

For $c = 0$, q' is strictly increasing on I .

Proof. To preserve a positive plateau, truncate and discretize the coefficient separately to the left and right of a closed subinterval of I , retain the value c on that subinterval, and impose a positive lower cutoff tending to zero and smaller than c . All choices are monotone and converge in L^1 . Increase the terminal smoothing parameter beyond the finite step bound. The convergence claims then follow from Lemmas 2.1 and 2.2. The products involved are uniformly bounded, and the spatial derivative bounds also give uniform continuity for evaluating them on the coupled paths.

For strict positivity, choose $b \in I$ later than the time in question and use (2.3). The function $h = \Phi_\gamma(b, \cdot)$ is smooth, convex, even, and differs from $|x|$ by a bounded function. Its derivative is therefore not constant. Twice differentiating the logarithm in (2.3) gives

$$\Phi_{xx}(t, x) = \mathbb{E}_{t,x}^{\text{tilt}}[h''] + c \text{Var}_{t,x}^{\text{tilt}}(h').$$

The tilted Gaussian law has a strictly positive density everywhere. The variance is strictly positive and the first term is nonnegative.

Inside I all functions are smooth in time as well as space, by the heat formula. Along X , the drifts of r , v , and w are respectively 0, $-v^2$, and $-3vw$, and their diffusion coefficients are v , w , and z . These identities follow by differentiating $r_t = -r_{xx}/2 - rr_x$. Itô's formula for v^2 and then for $w^2 - 2v^3$, with (2.12), proves (2.13)–(2.14). All differentiations under the expectation are justified by the bounded higher spatial derivatives on smaller intervals.

If $c = 0$, the same argument gives

$$(q')'(t) = \mathbb{E}[\Phi_{xxx}(t, X_t)^2] > 0.$$

To see strictness, start the drift-free interval at any earlier time in I . Its Gaussian transition kernel implies that X_t has a strictly positive density on $\mathbb{R}$. The function $\Phi_{xxx}(t, \cdot)$ cannot vanish identically: otherwise $\Phi(t, \cdot)$ would be an even quadratic polynomial, whereas it is one-Lipschitz and differs from $|x|$ by a bounded function. $\square$

3 Shape inequalities for finite step functions

On a positive constant interval, the curvature formula (2.14) contains the indefinite expression $z^2 - 12vw^2 + 6v^4$. The next section will make its expectation positive by adding a spatial derivative of zero expectation. Our task here is to prove the shape inequalities that make this possible: we compare the backward drift with a logarithmic derivative of the forward density.

We work with a fixed positive finite step coefficient and a smooth terminal condition. The estimates in this section may depend on these fixed data. Section 5 will pass only the resulting integrated inequality to general coefficients, using Lemma 2.1.

Fix a partition $0 = t_0 < t_1 < \dots < t_N = 1$, constants $0 < c_1 \leq \dots \leq c_N \leq M$, and $\gamma(t) = c_i$ on $[t_{i-1}, t_i]$. Let Φ solve

$$\Phi_t = -\frac{1}{2}(\Phi_{xx} + \gamma\Phi_x^2), \quad \Phi(1, x) = M^{-1} \log(2 \cosh(Mx)).$$

On a step where $\gamma = c$, put

$$l = c\Phi, \quad r = l_x, \quad v = r_x, \quad w = r_{xx}, \quad z = r_{xxx}.$$

Spatial derivatives of Φ agree across step boundaries; those of l are rescaled with c . Within a step, $h = e^l$ satisfies $h_t = -\frac{1}{2}h_{xx}$. In backward time τ this gives

$$r_\tau = \frac{1}{2}r_{xx} + rr_x. \tag{3.1}$$

Let $X_0 = 0$ and $dX_t = r(t, X_t)dt + dB_t$. The drift is bounded and spatially Lipschitz for these fixed data. Its density at positive times will be denoted by $f(t, x)$. Factor it as

$$f = e^l g, \quad s = -\partial_x \log g.$$

The factor e^l contains the backward solution. The remaining factor g evolves by the forward heat equation within each step. Indeed, the transition density from time a to time t in the same step is

$$p_{t-a}(x-y) \frac{h(t, x)}{h(a, y)}, \quad p_u(x) = (2\pi u)^{-1/2} e^{-x^2/(2u)}.$$

It integrates to one because $h(a, \cdot) = P_{t-a}h(t, \cdot)$, and differentiation verifies the forward equation with drift $r = h_x/h$. Consequently $g_t = \frac{1}{2}g_{xx}$. On the first step, $g(t, x) = p_t(x)/h(0, 0)$, so $s = x/t$. At a forward jump $c_+ - c_- = \delta \geq 0$, continuity of f gives

$$g_+ = g_- e^{-\delta\Phi}. \quad (3.2)$$

In particular g is positive and even, and s is odd.

The comparison we need has three parts. On the positive half-line the drift r is increasing and concave; its relative backward-time change r_τ/r is nonpositive and nondecreasing in x . The score s is also increasing and concave, and its ratio to r is nondecreasing in x . Lemmas 3.2 and 3.3 state these claims precisely. We first record the boundary estimates needed in their proofs.

3.1 Boundary estimates and comparison

A differentiated exponential error means that, for each fixed finite derivative order, the error and all its derivatives up to that order are $O(e^{-\kappa x})$ as $x \rightarrow +\infty$, for some $\kappa > 0$. Constants may change with the derivative order.

Lemma 3.1 (Differentiated tails). *On each closed backward step, uniformly in its time coordinate,*

$$l(t, x) = cx + C(t) + O(e^{-\kappa x}) \quad (x \rightarrow +\infty) \quad (3.3)$$

with differentiated exponential error. The functions are even, so the negative tail is determined by symmetry.

For the forward heat factor g , on every closed forward step with time bounded away from zero,

$$g(t, x) = \exp\left(-\frac{x^2}{2t} + \beta(t)x + C_0(t)\right) (1 + O(e^{-\kappa x})) \quad (x \rightarrow +\infty), \quad (3.4)$$

with differentiated relative error. In particular,

$$s(t, x) = \frac{x}{t} - \beta(t) + O(e^{-\kappa x}), \quad s_x(t, x) = \frac{1}{t} + O(e^{-\kappa x}), \quad s_{xx}(t, x) = O(e^{-\kappa x}). \quad (3.5)$$

The assertions include one-sided values at every step boundary.

The proof uses only Gaussian convolution and the finite-step jump rules; none of the shape inequalities proved below enters it.

Proof. Here is a Gaussian estimate that also controls differentiation. Suppose $q \in C^m(\mathbb{R})$ satisfies, for $0 \leq j \leq m$,

$$|q^{(j)}(y)| \leq Ae^{-\kappa y} \quad (y \geq 0), \quad |q^{(j)}(y)| \leq Ae^{K|y|} \quad (y < 0).$$

For $Y_x = \lambda x + b + \sqrt{V}Z$, where Z is standard normal, $0 < \lambda_* \leq \lambda \leq \lambda^*$, $|b| \leq B$, and $0 \leq V \leq V_*$,

$$\partial_x^j \mathbb{E}q(Y_x) = \lambda^j \mathbb{E}q^{(j)}(Y_x) = O(e^{-\kappa' x}) \quad (0 \leq j \leq m) \quad (3.6)$$

uniformly in these parameters, for some $\kappa' > 0$. Indeed, write $\mu = \lambda x + b$ and take x large enough that $\mu > 0$. On $Y_x \geq \mu/2$, use $Ae^{-\kappa\mu/2}$. On its complement, the global bound $|q^{(j)}(y)| \leq Ae^{K|y|}$, Cauchy–Schwarz, and $\mathbb{P}(Y_x < \mu/2) \leq e^{-\mu^2/(8V_*)}$ give a bound $C \exp(K\mu - \mu^2/(16V_*))$. This is smaller than an exponential in x ; the case $V = 0$ is immediate. Exponential integrability of the Gaussian justifies differentiation under the expectation. Adjusting the constant covers the remaining bounded range of x .

For backward evolution, suppose an even starting function l_0 has $l_0(x) = cx + C + \rho(x)$ with differentiated exponential error. Set $q(y) = e^{l_0(y) - cy - C} - 1$. This q obeys the hypotheses above: the positive tail follows by differentiating the exponential, and evenness of l_0 gives at most exponential growth for q and its derivatives on the negative tail. Gaussian tilting gives

$$P_\tau(e^{l_0})(x) = e^{cx + C + c^2\tau/2} (1 + \mathbb{E}q(x + c\tau + \sqrt{\tau}Z)).$$

Apply (3.6) uniformly for bounded τ and take the logarithm. Its relative error tends uniformly to zero, so differentiating $\log(1 + \cdot)$ proves (3.3) with all the stated derivatives. The initial condition starts this induction since, for $x > 0$,

$$\frac{c}{M} \log(2 \cosh Mx) = cx + \frac{c}{M} \log(1 + e^{-2Mx}).$$

At a backward boundary, l is multiplied by $c_-/c_+ > 0$, which preserves the required tail class. Finite induction proves the first claim.

Suppose at a positive time a that $g(a, y) = e^{-y^2/(2a) + \beta y + C}(1 + q(y))$ has the required tail class. Evenness of g implies the same exponential-growth bound for the derivatives of q on the negative tail. Completing the square gives

$$\begin{aligned} P_h g(a, \cdot)(x) &= \sqrt{\frac{a}{a+h}} \exp\left(-\frac{x^2}{2(a+h)} + \frac{a\beta}{a+h}x + C + \frac{\beta^2 ah}{2(a+h)}\right) \\ &\quad \times \left(1 + \mathbb{E}q\left(\frac{a}{a+h}x + \frac{\beta ah}{a+h} + \sqrt{\frac{ah}{a+h}}Z\right)\right). \end{aligned}$$

On a fixed interval $0 \leq h \leq H$, the coefficient of x in the Gaussian mean is at least $a/(a+H) > 0$. Thus (3.6) proves closure, uniformly including $h = 0$. At a jump, write $\Phi = x + D + \rho$ on the positive tail. Formula (3.2) replaces (β, C) by $(\beta - \delta, C - \delta D)$ and replaces $1 + q$ by $(1 + q)e^{-\delta\rho}$, preserving the differentiated relative error. Finite induction proves (3.4). Logarithmic differentiation then gives (3.5). $\square$

We will use the following elementary form of comparison. If on a compact space-time cylinder

$$u_t - \frac{1}{2}u_{xx} - bu_x - cu \geq 0, \quad c \leq C,$$

and $u \geq 0$ initially while $u \geq -\varepsilon$ on the lateral boundary, then

$$u(t, x) \geq -\varepsilon e^{(\max(C, 0) + 1)(t - t_0)}. \quad (3.7)$$

To see this, add the positive exponential on the right to u and multiply by $e^{-(\max(C,0)+1)(t-t_0)}$. The resulting zeroth-order coefficient is at most -1 . A negative space-time minimum, which cannot be on the parabolic boundary, has nonpositive time derivative, zero spatial gradient, and nonnegative second spatial derivative, contradicting its differential inequality. The same proof applies in a ball in $\mathbb{R}^d$ with $\frac{1}{2}\Delta$ in place of $\frac{1}{2}\partial_{xx}$. In particular, to exhaust the line or half-line it suffices that the lateral boundary errors tend uniformly to zero and that C is independent of the truncation. No uniform bound on the first-order coefficient b over the exhausting cylinders is needed.

3.2 Backward shape inequalities

Lemma 3.2 (Backward shapes). *For the fixed step data above, r is odd and $v > 0$. On $x > 0$,*

$$w \leq 0, \quad H := \frac{r_\tau}{r} = \frac{w}{2r} + v \leq 0, \quad H_x \geq 0. \quad (3.8)$$

The quotient defining H extends smoothly and evenly at zero.

Proof. Evenness is preserved by both heat evolution and rescaling. If $l_0'' > 0$, the logarithm of its heat evolution obeys

$$\partial_{xx} \log P_\tau e^{l_0}(x) = \mathbb{E}_\nu l_0''(Y) + \text{Var}_\nu(l_0'(Y)) > 0,$$

where ν is the Gaussian law of $Y = x + \sqrt{\tau}Z$ tilted by $e^{l_0(Y)}$; at $\tau = 0$ use $l_0''(x) > 0$ directly. Since the terminal second derivative is $cM \text{sech}^2(Mx) > 0$ and backward rescalings are positive, this proves $v > 0$. Thus $r(x) > 0$ for $x > 0$. All required derivatives are bounded on closed steps, by smoothness on compact sets and Lemma 3.1 at infinity.

Differentiating (3.1) twice gives

$$w_\tau = \frac{1}{2}w_{xx} + rw_x + 3vw.$$

The terminal $w = -2cM^2 \text{sech}^2(Mx) \tanh(Mx)$ is nonpositive for $x > 0$. The lateral values are $w(\tau, 0) = 0$ and $w(\tau, A) = o_A(1)$ uniformly on a closed step. Apply (3.7) to $-w$, with $C = 3 \sup v$, and let $A \rightarrow \infty$. Backward jumps multiply w by a positive constant, so the sign holds throughout.

Because $r(x) = v(0)x + O(x^3)$ with $v(0) > 0$, odd smoothness makes w/r smooth and even at zero. Direct differentiation of $r_\tau = rH$ gives

$$H_\tau = \frac{1}{2}H_{xx} + (v/r + r)H_x + vH, \quad (3.9)$$

$$(H_x)_\tau = \frac{1}{2}(H_x)_{xx} + (v/r + r)(H_x)_x + ((v/r)_x + 2v)H_x + wH. \quad (3.10)$$

At the terminal endpoint of the last step,

$$H = (cM - M^2) \text{sech}^2(Mx) \leq 0, \quad H_x \geq 0 \quad (x > 0).$$

For (3.9), write $v/r + r = x^{-1} + \eta(x)$, where η is smooth odd and $\eta(x) = O(x)$ near zero. The radial lift $\hat{H}(y) = H(|y|)$ in $\mathbb{R}^3$ satisfies

$$\hat{H}_\tau = \frac{1}{2}\Delta \hat{H} + \eta(|y|) \frac{y}{|y|} \cdot \nabla \hat{H} + v(|y|)\hat{H}.$$

Both the lifted function and vector field are smooth at zero: a smooth even function lifts smoothly, and $\eta(x)/x$ is smooth even. Since $H \rightarrow 0$ uniformly at infinity, comparison on balls followed by exhaustion gives $H \leq 0$.

For H_x , use intervals $[\varepsilon, A]$. The source wH is nonnegative, and

$$(v/r)_x = \frac{w}{r} - \frac{v^2}{r^2} < 0, \quad (v/r)_x + 2v \leq 2 \sup v.$$

The lateral errors are $H_x(\tau, \varepsilon) = O(\varepsilon)$, uniformly by smooth evenness, and $H_x(\tau, A) = o_A(1)$ by the tails. Formula (3.7), with a bound independent of ε, A , proves $H_x \geq 0$ on taking the two limits.

Finally, at a backward jump write $r_{\text{new}} = \alpha r_{\text{old}}$, where $0 < \alpha \leq 1$. Its derivatives rescale identically, and hence

$$H_{\text{new}} = H_{\text{old}} - (1 - \alpha)v_{\text{old}}, \quad (H_x)_{\text{new}} = (H_x)_{\text{old}} - (1 - \alpha)w_{\text{old}}.$$

These preserve both signs. Induction over the finitely many steps completes the proof. $\square$

3.3 Forward score inequalities

The logarithmic derivative $s = -g_x/g$ measures the part of the density not contained in e^l . We need both its concavity and a comparison with r , rather than concavity of either function alone. The density transform and its use on constant intervals also appear in Lopatto's positive-temperature analysis [13, Sections 3, 7, and 9]. Here we prove the additional ratio sign needed by the coercive identity.

Lemma 3.3 (Forward score and ratio). *For every $t > 0$, s is odd and, with $a = s_x$,*

$$a \geq \frac{1}{t} > 0, \quad s_{xx} \leq 0 \quad (x > 0), \quad j := a - \frac{s}{r}v = r\partial_x(s/r) \geq 0 \quad (x \in \mathbb{R}). \quad (3.11)$$

The ratios and j extend smoothly at zero. At every forward step boundary, j is unchanged.

Proof. Evenness and positivity of g give odd smoothness of s . Within a step, the heat equation for g gives

$$s_t = \frac{1}{2}s_{xx} - ss_x, \quad (3.12)$$

$$a_t = \frac{1}{2}a_{xx} - sa_x - a^2, \quad (3.13)$$

$$(s_{xx})_t = \frac{1}{2}(s_{xx})_{xx} - s(s_{xx})_x - 3as_{xx}. \quad (3.14)$$

On the first interval $s = x/t$, so the first two claims hold explicitly. At a jump of size $\delta = c_+ - c_-$, (3.2) gives

$$s_+ = s_- + \delta\Phi_x, \quad a_+ = a_- + \delta\Phi_{xx}, \quad (s_{xx})_+ = (s_{xx})_- + \delta\Phi_{xxx}. \quad (3.15)$$

Here $\Phi_{xx} > 0$ and $\Phi_{xxx} \leq 0$ on the positive half-line by Lemma 3.2.

For subsequent intervals, put $u = a - 1/t$. Equation (3.13) becomes the linear equation

$$u_t = \frac{1}{2}u_{xx} - su_x - (a + 1/t)u.$$

For the fixed smooth solution, a is bounded on a closed step; this follows from (3.5) and compactness and does not presume its sign. Thus the zeroth-order coefficient is bounded above independently of a spatial truncation. Initially $u \geq 0$, by the preceding step and (3.15); at both infinities $u \rightarrow 0$ uniformly by Lemma 3.1. Comparison on $[-A, A]$ and exhaustion prove $a \geq 1/t$. Next apply comparison to $-s_{xx}$ on $[0, A]$ using (3.14). Its zeroth coefficient is $-3a \leq 0$, its initial values are nonnegative, and its boundary values tend to zero, at zero by oddness and at infinity by (3.5). This proves $s_{xx} \leq 0$.

For the ratio inequality, first note that $s(x) > 0$ and $r(x) > 0$ for $x > 0$. On this half-line define

$$d_s = \frac{s_x}{s}, \quad d_r = \frac{r_x}{r}, \quad \rho = d_s - d_r.$$

Then $j = s\rho$, so it suffices to prove $\rho \geq 0$. On the first step, $s = x/t$ and the concavity of r gives $r(x) = \int_0^x v(y)dy \geq xv(x)$; hence $\rho = 1/x - v/r \geq 0$. Within any later step, forward time reverses (3.1), so $r_t = -rH$. Consequently

$$(d_s)_t = \frac{1}{2}(d_s)_{xx} + d_s(d_s)_x - s_{xx}, \quad (d_r)_t = -H_x.$$

Also $\frac{1}{2}(d_r)_{xx} + d_r(d_r)_x = \frac{1}{2}(w/r)_x = H_x - w$. Subtracting gives

$$\rho_t = \frac{1}{2}\rho_{xx} + d_s\rho_x + (d_r)_x\rho + 2H_x - w - s_{xx}. \quad (3.16)$$

The source is nonnegative by the signs already proved, while $(d_r)_x = w/r - v^2/r^2 < 0$.

Here are the endpoint details for this last comparison. On a closed step away from time zero, $v(t, 0) > 0$ is continuous and bounded below, and $a(t, 0) \geq 1/t > 0$. Odd Taylor expansions therefore give, uniformly,

$$d_s = x^{-1} + O(x), \quad d_r = x^{-1} + O(x), \quad \rho = O(x) \quad (x \downarrow 0).$$

At infinity, (3.3) and (3.5) give $d_r = O(e^{-\kappa x})$ and $d_s = (x - \beta(t)t)^{-1} + O(e^{-\kappa'x})$, so $\rho \rightarrow 0$ uniformly. Use (3.7) on $[\varepsilon, A]$, with upper zeroth-order bound zero, and let $\varepsilon \downarrow 0$, $A \rightarrow \infty$. This proves $\rho \geq 0$ provided it is nonnegative at the start of the step.

To verify that condition, write at a forward jump

$$r_+ = \alpha r_-, \quad v_+ = \alpha v_-, \quad s_+ = s_- + \lambda r_-, \quad a_+ = a_- + \lambda v_-, \quad \alpha = \frac{c_+}{c_-} > 0, \quad \lambda = \frac{c_+ - c_-}{c_-} \geq 0.$$

Then direct cancellation gives

$$j_+ = a_- + \lambda v_- - \frac{s_- + \lambda r_-}{\alpha r_-} \alpha v_- = j_-.$$

Since $\rho_+ = j_+/s_+$ for $x > 0$, its initial sign is preserved. This proves the ratio claim by induction. Smoothness of s/r at zero follows by factoring x from both odd functions; in particular j is even and $j(0) = 0$. Reflection gives the assertion on the whole line. $\square$

3.4 Variables for the integral certificate

We now express the shape inequalities as nonnegative quantities suited to polynomial algebra. Set

$$R = r^2, \quad K = rs, \quad b = -w/r, \quad j = s_x - (s/r)v = r(s/r)_x.$$

The first two quantities are nonnegative by the common signs of r and s , and $b \geq 0$ expresses the concavity of r . The ratio comparison is precisely $j \geq 0$. The remaining three quantities encode the backward-time and forward-concavity inequalities:

$$B = b - 2v = -2H, \quad S = z + bv - 2Rb = 2rH_x, \quad L = -rs_{xx}.$$

All are nonnegative. The following lemma collects their derivative rules and the integration-by-parts identity that will be used in the coercive calculation.

Lemma 3.4 (Certificate signs and integration by parts). *At a fixed positive time in a constant step, the functions R, K, v, b, j, L, B, S defined above are even and extend smoothly across zero. They satisfy*

$$R, K, v, j, L, B, S \geq 0.$$

Writing $\mathcal{D} = r\partial_x$, one has

$$\mathcal{D}(R, K, v, b, j) = (2Rv, 2vK + Rj, -Rb, -z - bv, -jv + Kb - L). \quad (3.17)$$

For every polynomial F in R, K, v, b, j ,

$$\mathbb{E}_f[\mathcal{D}F + (v + R - K)F] = 0. \quad (3.18)$$

Every term in this identity is absolutely integrable.

Proof. The signs of R, K, v, j, L follow from Lemmas 3.2 and 3.3; for K use the common sign of the odd functions r, s . Since $H = v - b/2$, we have $B = -2H \geq 0$. On $x > 0$, differentiation gives

$$S = 2rH_x \geq 0.$$

All these expressions are even. The ratios w/r and s/r are smooth at zero since their numerators and denominator are odd and $r_x(0) > 0$. The inequalities consequently extend there by continuity. For example $b(0) = -z(0)/v(0)$, $j(0) = 0$, and $S(0) = 0$.

The derivative rules follow directly from the definitions. The only one involving the forward score is

$$rj_x = rs_{xx} - av - sw + \frac{sv^2}{r} = -jv + Kb - L,$$

which gives the last rule in (3.17).

The density has logarithmic derivative $f_x/f = r - s$. Therefore

$$\partial_x(rFf) = (\mathcal{D}F + (v + r(r - s))F)f = (\mathcal{D}F + (v + R - K)F)f.$$

By the differentiated tails, r is bounded and tends to c at the positive tail, v, w, z, b and their derivatives decay exponentially, $s = x/t + O(1)$, $j = 1/t + O((1+x)e^{-\kappa x})$, and $K = O(1+x)$. Every polynomial F and its differentiated expression thus have at most polynomial growth. Moreover

$$f(t, x) \leq C \exp\left(-\frac{x^2}{2t} + C|x|\right).$$

These bounds prove absolute integrability and $rFf \rightarrow 0$ at both infinities. Integrating the displayed derivative on $[-A, A]$ and letting $A \rightarrow \infty$ proves (3.18). $\square$

4 An exact coercive identity

We retain the fixed positive time, positive constant step, and notation of Lemma 3.4. We will prove

$$\mathbb{E}_f[z^2 - 12vw^2 + 6v^4] \geq \frac{1}{1000} \mathbb{E}_f v^4.$$

The expression need not be positive pointwise. Subtracting a spatial derivative with zero expectation preserves its expectation; the calculation below decomposes the resulting expression into nonnegative terms. The variables introduced at the end of the previous section satisfy

$$R, K, v, B, j, L, S \geq 0, \quad b = B + 2v \geq 0, \quad Rb^2 = w^2. \quad (4.1)$$

In particular, the signs of B and S record the backward inequalities $H \leq 0$ and $H_x \geq 0$.

For a polynomial F in R, K, v, b, j , set

$$\mathcal{D}F = r\partial_x F, \quad \mathcal{W}(F) = \mathcal{D}F + (v + R - K)F. \quad (4.2)$$

The derivative rules in (3.17) determine $\mathcal{W}$ algebraically. Its analytic meaning is

$$f\mathcal{W}(F) = \partial_x(rfF), \quad \mathbb{E}_f\mathcal{W}(F) = 0. \quad (4.3)$$

Lemma 3.4 supplies the integrability and vanishing boundary terms for every polynomial used here. Thus it suffices to find F for which subtracting $\mathcal{W}(F)$ from the curvature expression leaves a positive multiple of v^4 and other nonnegative terms.

4.1 Choosing the derivative correction

We first arrange the terms containing z . Set

$$(A, C, k, T_0, g_0, h) = \left(\frac{1}{13}, \frac{53}{24}, \frac{19}{16}, \frac{73}{48}, \frac{2331}{10000}, \frac{87}{2500} \right) \quad (4.4)$$

and define

$$Q = Avb + Cv^2 + kKv - 2kRb + T_0Rv, \quad T = g_0Rv + h jv. \quad (4.5)$$

The terms $(z + Q)^2$ and ST will be nonnegative: the first is a square, and $S, T \geq 0$. Since $S = z + bv - 2Rb$, their terms linear in z have total coefficient $2Q + T$.

The only derivative rule containing z is $\mathcal{D}b = -z - bv$. Consequently the coefficient of z in $\mathcal{W}(F)$ is $-\partial_b F$. We therefore choose $F = F_2$ so that $\partial_b F_2 = 2Q + T$. Set

$$(x_0, y, m, n_0) = \left(\frac{4663}{5000}, \frac{6853}{10000}, -\frac{1051}{400}, -\frac{1073}{500} \right). \quad (4.6)$$

The following polynomial has the required derivative:

$$\begin{aligned} F_2 = & (Av - 2kR)b^2 \\ & + b(2Cv^2 + 2kKv + (g_0 + 2T_0)Rv + h jv) \\ & + x_0v^2j + yv^3 + mKv^2 + n_0Rv^2. \end{aligned} \quad (4.7)$$

Its first two lines are the required antiderivative in b ; the last line is independent of b and will be used in the remaining terms. Directly,

$$\partial_b F_2 = 2Avb - 4kRb + 2Cv^2 + 2kKv + (g_0 + 2T_0)Rv + h jv = 2Q + T.$$

It follows that

$$z^2 - 12Rvb^2 + 6v^4 - (z + Q)^2 - ST - \mathcal{W}(F_2)$$

has no z dependence. This is the reason for the choice of the derivative correction.

The remaining calculation uses one further favorable term from the forward score inequality. Let $\mathcal{W}_0$ be the polynomial rule obtained from $\mathcal{W}$ by omitting $-L$ in $\mathcal{D}j = -jv + Kb - L$. Since

$$\partial_j F_2 = x_0v^2 + h bv, \quad \mathcal{W}(F_2) = \mathcal{W}_0(F_2) - L(x_0v^2 + h bv), \quad (4.8)$$

replacing $\mathcal{W}$ by $\mathcal{W}_0$ leaves the nonnegative term $L(x_0v^2 + h bv)$ to be retained in the final identity. We now have only a polynomial in R, K, v, b, j to decompose.

4.2 The nonnegative remainder

Define the five polynomials

$$\begin{aligned}
P_0 &= Rj(b-7v)^2, \\
P_1 &= RB\left(b-\frac{7}{3}v\right)^2, \\
P_2 &= Kv\left(b-\frac{9}{2}v\right)^2, \\
P_3 &= vB\left(K+\frac{1}{4}R-\frac{1}{7}v\right)^2, \\
P_4 &= \left(vb-\frac{14}{5}v^2-\frac{13}{4}Kv+\frac{4}{3}Rv\right)^2,
\end{aligned} \tag{4.9}$$

with positive coefficients

$$(p_0, p_1, p_2, p_3, p_4) = \left(\frac{337}{10000}, \frac{183}{2500}, \frac{411}{10000}, \frac{11461}{5000}, \frac{647}{10000}\right). \tag{4.10}$$

Let P_{rem} be the polynomial in R, K, v, B, j whose nonzero coefficients are listed in Table 1. All its variables and coefficients are nonnegative.

Lemma 4.1 (Rational polynomial identity). *With these definitions, $B = b-2v$, and $S = z+bv-2Rb$, one has the polynomial identity over $\mathbb{Q}$*

$$\begin{aligned}
z^2 - 12Rvb^2 + 6v^4 &= (z+Q)^2 + ST + \sum_{i=0}^4 p_i P_i + \mathcal{W}(F_2) \\
&\quad + L(x_0 v^2 + h b v) + P_{\text{rem}}.
\end{aligned} \tag{4.11}$$

Proof. Equation (4.8) reduces the assertion to the same identity with $\mathcal{W}_0$ and without the displayed L term. The construction of F_2 above cancels every term containing z . Expand the residual

$$z^2 - 12Rvb^2 + 6v^4 - (z+Q)^2 - ST - \mathcal{W}_0(F_2) - \sum_{i=0}^4 p_i P_i$$

using (3.17), substitute $b = B + 2v$, and collect monomials in R, K, v, B, j . The coefficients are exactly those of Table 1, and every unlisted coefficient is zero. This proves the identity by rational arithmetic. $\square$

Proposition 4.2 (Integrated coercivity). *Suppose r, v, w, z, s and the variables R, K, b, j, L, B, S are related as in Lemma 3.4, with the derivative rules (3.17) and signs (4.1). Assume their quotients extend across zero. Let f be a probability density satisfying $f_x/f = r - s$ such that every term of (4.11) is integrable and rfF_2 tends to zero at both spatial infinities. Then*

$$\mathbb{E}_f[z^2 - 12vw^2 + 6v^4] \geq \frac{1661507}{1521000000} \mathbb{E}_f v^4 \geq \frac{1}{1000} \mathbb{E}_f v^4. \tag{4.12}$$

Proof. Integrating (4.11) removes $\mathcal{W}(F_2)$ by (4.3). Every remaining term on the right is nonnegative. Indeed, the prefactors of the five squares P_i are $Rj, RB, Kv, vB, 1$; $T = v(g_0 R + h j) \geq 0$; and $L(x_0 v^2 + h b v) \geq 0$. The table gives the stronger bound

$$P_{\text{rem}} \geq \frac{1661507}{1521000000} v^4, \quad \frac{1661507}{1521000000} - \frac{1}{1000} = \frac{140507}{1521000000} > 0.$$

Using $Rb^2 = w^2$ proves (4.12). $\square$

Monomial	Coefficient	Monomial	Coefficient
R^2v^2	223/160000	R^2vB	4007/240000
R^2B^2	1097/120000	RKv^2	83/80000
$RKvB$	3343/24000	Rv^3	16259/23400000
Rv^2B	30179/5460000	Rv^2j	26/625
RvB^2	107/15000	$RvBj$	3/2500
RB^3	121/32500	RB^2j	11/10000
K^2v^2	579/20000	K^2vB	207/2500
Kv^3	911/975000	Kv^2B	349/420000
Kv^2j	5011/5000	KvB^2	133/130000
$KvBj$	87/2500	v^4	1661507/1521000000
v^3B	633799/621075000	v^2B^2	10657/1690000

Table 1: All 22 nonzero coefficients of P_{rem} ; every unlisted coefficient is zero. Each listed coefficient is positive; the coefficient of v^4 gives the coercive lower bound.

For the smooth finite-step data of Section 3, Lemma 3.4 verifies all the hypotheses of this proposition. Although the proof uses the density score and its ratios, the resulting inequality involves only v, w, z . This is the form that passes to an integrable order parameter.

5 Strict curvature on constant intervals

We now pass from the smooth finite-step calculation to the order parameters in $\mathcal{U}$. Both sides of the final integrated inequality depend only on spatial derivatives of Φ through order four. The compact-strip bounds and convergence from Section 2 therefore suffice to pass to the limit; no uniform tail constants or convergence of the forward scores are needed.

Proposition 5.1 (Constant-interval curvature). *Let $\gamma \in \mathcal{U}$ and suppose $\gamma = c$ on an open interval $I \subset (0, 1)$. For its controlled diffusion, put $q(t) = \mathbb{E}\Phi_{\gamma,x}(t, X_t)^2$. If $c > 0$, then q' is strictly convex on I ; more precisely, with $v = c\Phi_{\gamma,xx}$,*

$$q'''(t) \geq \frac{1}{1000c^2} \mathbb{E}v(t, X_t)^4 > 0, \quad t \in I. \quad (5.1)$$

If $c = 0$, then q' is strictly increasing on I .

Proof. Suppose first $c > 0$ and fix $t \in I$. Choose a closed neighborhood of t inside I . Lemma 2.3 supplies positive nondecreasing finite-step approximants γ_n , equal to c there, and terminal smoothings $M_n^{-1} \log(2 \cosh M_n x)$ with $M_n \geq \|\gamma_n\|_\infty$ and $M_n \rightarrow \infty$. Write X^n for the associated diffusions and $r_n = c\Phi_{n,x}$, $v_n = r_{n,x}$, $w_n = r_{n,xx}$, $z_n = r_{n,xxx}$ on that neighborhood.

Lemmas 3.2, 3.3, and 3.4 verify all sign, integrability, and boundary hypotheses of Proposition 4.2. Therefore

$$\mathbb{E}[z_n^2 - 12v_n w_n^2 + 6v_n^4](t, X_t^n) \geq \frac{1}{1000} \mathbb{E}v_n(t, X_t^n)^4.$$

The derivatives through order four converge uniformly in space and are uniformly bounded, by Lemma 2.1. The coupled paths X_t^n converge, and the next spatial derivative bounds ensure uniform continuity when evaluating at these paths. Thus each expectation in the display converges to its counterpart for γ . In particular no limit of s_n , j_n , or w_n/r_n is needed.

Equation (2.14) identifies the resulting left side with $c^2 q'''(t)$. Lemma 2.3 also gives $v(t, x) = c\Phi_{\gamma, xx}(t, x) > 0$ for every x , so its fourth moment is strictly positive. This proves (5.1) at every interior time, hence strict convexity of q' . The assertion for $c = 0$ is the last part of Lemma 2.3; it is not obtained by dividing by c . $\square$

6 Full support and regularity

We now turn the constant-interval curvature into full support. The first variation must allow removal of arbitrary Stieltjes mass, not only removal of atoms: a minimizer need not initially be known to have a discrete or absolutely continuous measure. For related variational characterizations see Chen, Handschy, and Lerman [8, Propositions 3–4] and Jagannath and Tobasco [12, Lemma 3.2 and Corollary 3.6].

Lemma 6.1 (First variation in every admissible integrable direction). *Let $\gamma \in \mathcal{U}$, and suppose that $\eta \in L^1(0, 1)$ has the property that $\gamma + \epsilon\eta \in \mathcal{U}$ for $0 \leq \epsilon \leq \epsilon_0$, where $\epsilon_0 > 0$. With X and q from Lemma 2.2,*

$$\left. \frac{d}{d\epsilon} \right|_{\epsilon=0+} \mathcal{P}(\gamma + \epsilon\eta) = \frac{1}{2} \int_0^1 \eta(t)(q(t) - t) dt. \quad (6.1)$$

Proof. Put $\beta = \gamma + \epsilon\eta$, $u = \Phi_{\gamma, x}$, $\tilde{u} = \Phi_{\beta, x}$, and $D = \Phi_\beta - \Phi_\gamma$. Subtracting the two equations gives

$$D_t = -\frac{1}{2}D_{xx} - \frac{1}{2}\beta(\tilde{u} + u)D_x - \frac{1}{2}(\beta - \gamma)u^2.$$

Let $Y_0^\epsilon = 0$ and

$$dY_t^\epsilon = \frac{1}{2}\beta(t)(\tilde{u} + u)(t, Y_t^\epsilon) dt + dB_t.$$

This diffusion exists uniquely by Lemma 2.1; its drift is bounded by the integrable function β . Itô's formula on $[0, T]$ yields the following identity. One may justify it without a classical time derivative by approximating γ and β separately by nondecreasing step functions, using the same smooth terminal datum for the pair. The two solutions, their spatial derivatives, and the associated linear diffusions converge on $[0, T]$ by Lemma 2.1 and the same Gronwall estimate as in Lemma 2.2. The source converges in L^1 , since the gradients are bounded by one. Passing to the limit in their classical identities gives

$$D(0, 0) = \mathbb{E}D(T, Y_T^\epsilon) + \frac{1}{2} \int_0^T (\beta - \gamma)(t) \mathbb{E}[u(t, Y_t^\epsilon)^2] dt.$$

The time-local form of (2.5) gives $\|D(T, \cdot)\|_\infty \leq \frac{3}{2} \int_T^1 |\beta - \gamma|$. Letting $T \uparrow 1$ therefore proves the exact identity

$$\frac{\Phi_\beta(0, 0) - \Phi_\gamma(0, 0)}{\epsilon} = \frac{1}{2} \int_0^1 \eta(t) \mathbb{E}[u(t, Y_t^\epsilon)^2] dt. \quad (6.2)$$

The coefficients β have uniformly bounded L^1 norms for $0 \leq \epsilon \leq \epsilon_0$, and converge to γ in L^1 . Lemma 2.1 gives uniform convergence $\tilde{u} \rightarrow u$ and common spatial Lipschitz bounds on every compact strip. Coupling the diffusions with the same Brownian motion and using Gronwall's inequality gives $Y^\epsilon \rightarrow X$ uniformly there. The integrand in (6.2) consequently converges for each $t < 1$ to $\eta(t)q(t)$ and is bounded in absolute value by $|\eta(t)|$. Dominated convergence and differentiation of the linear penalty term in $\mathcal{P}$ prove (6.1). $\square$

Lemma 6.2 (Contact conditions). *Let γ minimize $\mathcal{P}$ over $\mathcal{U}$, and let μ be its locally finite Stieltjes measure on $[0, 1)$, with the convention $\mu(\{0\}) = \gamma(0)$. Thus $\gamma(t) = \mu([0, t])$. Define*

$$G(s) = \int_s^1 (q(t) - t) dt, \quad S = \text{supp } \mu \subset [0, 1). \quad (6.3)$$

Then $G \geq 0$ on $[0, 1)$ and $G = 0$ on S . At every $s \in S \cap (0, 1)$,

$$q(s) = s, \quad q'(s) \leq 1. \quad (6.4)$$

If $0 \in S$, then also $q'(0) \leq 1$, with the derivative from the right.

Proof. For $s < 1$, the direction $\eta = \mathbf{1}_{[s, 1]}$ adds an atom to μ and is admissible. Minimality and Lemma 6.1 give $G(s) \geq 0$. If $K \subset [0, 1)$ is compact, let $\nu = \mu|_K$ and take $\eta_K(t) = -\nu([0, t])$. This bounded direction is admissible for $0 \leq \epsilon \leq 1$, since it replaces μ by $\mu - \epsilon\nu$. Fubini's theorem gives

$$0 \leq D\mathcal{P}(\gamma)[\eta_K] = -\frac{1}{2} \int_K G(s) \mu(ds).$$

The interchange is legitimate because $|q(t) - t| \leq 1$ and $\nu(K) < \infty$. Since $G \geq 0$, it vanishes μ -almost everywhere on every such K . Its continuity then gives $G = 0$ on the relative support S .

By Lemma 2.2, $G'(s) = s - q(s)$ and $G''(s) = 1 - q'(s)$ on compact strips. Every interior support point is a minimum of G , proving (6.4). If $0 \in S$, use $G(0) = 0$ and $q(0) = 0$ to write

$$G(h) = \frac{1}{2}(1 - q'(0))h^2 + o(h^2).$$

Nonnegativity for $h > 0$ proves the asserted one-sided condition. $\square$

Lemma 6.3 (No terminal plateau). *For a minimizer γ , there is no $a < 1$ and finite $c \geq 0$ such that $\gamma = c$ on $(a, 1)$.*

Proof. The endpoint conclusion is recorded by Auffinger, Chen, and Zeng [4, Remark 7]. We give the quantitative argument needed here, including arbitrary positive constant plateaus. The boundary-layer method is also related to the zero-terminal-segment argument in El Alaoui, Montanari, and Sellke [9, Lemma 6.12 and the proof of Theorem 5 in the cited preprint]; their positivity-support statement concerns a different admissible class. Suppose such a plateau exists. We show that for some $C > 0$,

$$1 - q(t) \geq C\sqrt{1 - t} \quad \text{for all } t \text{ sufficiently close to one.} \quad (6.5)$$

Fix $s \in (a, 1)$, and let ρ_s denote the law of X_s . Choose $R < \infty$ with $p_0 = \rho_s([-R, R]) > 0$; no density assumption on ρ_s is needed.

When $c > 0$, set $h(t, y) = \mathbb{E} \exp(c|y + \sqrt{1 - t}Z|)$, with Z standard normal. The heat formula gives $\Phi(t, y) = c^{-1} \log h(t, y)$ and drift $c\Phi_x = h_x/h$. Its transition density on the plateau is

$$k_{s,t}(x, y) = p_{t-s}(y - x) \frac{h(t, y)}{h(s, x)}, \quad p_v(z) = \frac{e^{-z^2/(2v)}}{\sqrt{2\pi v}}. \quad (6.6)$$

Indeed, the heat-semigroup identity makes $h(t, B_t)/h(s, B_s)$ a positive mean-one martingale when Brownian motion is started at (s, x) . Its stochastic logarithm has the bounded coefficient $h_x/h = c\Phi_x$, so change of measure gives precisely this drift and kernel. For $c = 0$ use the same kernel formula with $h \equiv 1$.

Put $L = 1 - s$ and $\delta = L/2$. Uniformly for $s + \delta \leq t < 1$, $|x| \leq R$, and $|y| \leq 1$,

$$p_{t-s}(y - x) \geq (2\pi L)^{-1/2} e^{-(R+1)^2/(2\delta)} =: k_* > 0,$$

while $h(t, y) \geq 1$ and $h(s, x) \leq e^{cR} \mathbb{E} e^{c\sqrt{L}|Z|} =: M$. Consequently the mixture of (6.6) against ρ_s is a density f_t satisfying

$$f_t(y) \geq p_0 k_*/M =: m > 0 \quad (s + \delta \leq t < 1, |y| \leq 1). \quad (6.7)$$

For $\epsilon = 1 - t$ and $|y| \leq \sqrt{\epsilon}$, differentiation of the terminal heat formula gives, also when $c = 0$,

$$u(t, y) = \frac{\mathbb{E}[\operatorname{sgn}(y + \sqrt{\epsilon}Z) e^{c|y + \sqrt{\epsilon}Z|}]}{\mathbb{E} e^{c|y + \sqrt{\epsilon}Z|}}.$$

For $\epsilon \leq 1$ the denominator is at most $D = e^c \mathbb{E} e^{c|Z|} < \infty$. The unnormalized mass of each sign is at least $p_* := \Pr(Z \geq 1) > 0$, because $|y/\sqrt{\epsilon}| \leq 1$ and the exponential tilt is at least one. Each sign thus has probability at least $\theta = p_*/D \in (0, 1/2)$ under the normalized tilt. Hence $|u(t, y)| \leq 1 - 2\theta$ and $1 - u(t, y)^2 \geq 4\theta(1 - \theta) =: \kappa > 0$. Integrating this bound over $[-\sqrt{\epsilon}, \sqrt{\epsilon}]$ and using (6.7) proves (6.5) with $C = 2m\kappa$.

It follows that $q(t) - t \leq (1 - t) - C\sqrt{1 - t} < 0$ sufficiently near one. Thus $G(s') < 0$ for s' sufficiently near one, contrary to Lemma 6.2. This also covers $c = 0$. $\square$

Proof of Theorem 1.1. Let S be the support in Lemma 6.2. Since S is relatively closed in $[0, 1)$, a nonempty component of its complement is an initial, interior, or terminal interval. On each such interval γ is constant. A terminal component is impossible by Lemma 6.3; that lemma also excludes empty support, for which $\gamma = 0$.

Consider a remaining gap with endpoints $0 \leq a < b < 1$ and with $a, b \in S$. By Lemma 6.2, $q(a) = a$, $q(b) = b$, and $q'(a), q'(b) \leq 1$, including the one-sided interpretation at $a = 0$. If its constant coefficient is positive, Proposition 5.1 makes q' strictly convex inside the gap. By continuity up to its endpoints,

$$q'(t) < \frac{b - t}{b - a} q'(a) + \frac{t - a}{b - a} q'(b) \leq 1, \quad a < t < b.$$

Integrating contradicts $q(b) - q(a) = b - a$, as illustrated by the positive area deficit in Figure 1. If the constant coefficient is zero, the strict increase in Proposition 5.1 instead gives $q'(t) < q'(b) \leq 1$, with the same contradiction.

It remains only to consider an initial component $[0, b)$ when $0 \notin S$ and $b \in S$. Here $\gamma = 0$ on the gap. Strict increase gives $q'(t) < q'(b) \leq 1$ for $0 < t < b$, whereas $q(0) = 0$ and $q(b) = b$. Integration is again a contradiction. Thus $S = [0, 1)$. The contact identities give $q(t) = t$ for $0 < t < 1$, and the identity at zero follows from symmetry. Differentiating this identity using Lemma 2.2 gives $\mathbb{E} \Phi_{\gamma, xx}(t, X_t)^2 = 1$, including the right derivative at zero. Finally, every nonempty interval (a, b) with $0 \leq a < b < 1$ has positive μ mass, so $\gamma(b) - \gamma(a) = \mu((a, b]) > 0$. $\square$

6.1 Smoothness and positivity of the density

The smoothness conclusion and its deduction from saturated consistency appear in Chen [7, Theorem 1.3 and Proposition 4.6]. We include the argument to show how Theorem 1.1 determines the regularity of the minimizing order parameter. The polynomial-moment differentiation follows Auffinger and Chen [1, Lemma 2 and the proof of Theorem 2(ii)], in its zero-temperature form in Chen [7, Appendix B.6]; the Hessian semimartingale identity is also given by Chen, Handschy, and Lerman [8, Lemma 3]. We then use the coercive inequality of Section 4 to prove that the density is strictly positive at every positive time below one.

Corollary 6.4 (Smooth positive density of the minimizing measure). *Let $\gamma \in \mathcal{U}$ minimize $\mathcal{P}$, and let X be its controlled diffusion. Then $\gamma(0) = 0$ and γ is smooth on $[0, 1)$, with derivatives at zero understood from the right. Moreover,*

$$\gamma(t) = \frac{\mathbb{E}[\Phi_{\gamma,xxx}(t, X_t)^2]}{2\mathbb{E}[\Phi_{\gamma,xx}(t, X_t)^3]}, \quad 0 \leq t < 1. \quad (6.8)$$

Consequently $\mu_\gamma(dt) = \gamma'(t) dt$ on $[0, 1)$, where γ' is smooth and nonnegative. In fact, for $0 < t < 1$,

$$2\gamma'(t)\mathbb{E}[\Phi_{\gamma,xx}(t, X_t)^3] \geq \frac{\gamma(t)^2}{1000}\mathbb{E}[\Phi_{\gamma,xx}(t, X_t)^4] > 0. \quad (6.9)$$

In particular $\gamma'(t) > 0$ on $(0, 1)$.

Proof. Write $f_k = \partial_x^k \Phi_\gamma$ for $k \geq 1$, and evaluate these functions along X unless otherwise indicated. All spatial derivatives are bounded on each compact time strip by Lemma 2.1. Differentiating the PDE and applying Itô's formula gives

$$df_k(t, X_t) = -\frac{\gamma(t)}{2} \sum_{j=1}^{k-1} \binom{k}{j} f_{j+1}(t, X_t) f_{k-j+1}(t, X_t) dt + f_{k+1}(t, X_t) dB_t. \quad (6.10)$$

The sum is empty for $k = 1$. For general integrable γ , this identity follows by the finite-step approximation and Itô isometry used in Lemma 2.2. On a fixed strip the spatial factors converge uniformly along the coupled paths, the time coefficients converge in L^1 , and the stochastic integrals converge in L^2 .

For $a = f_2$ and $b = f_3$, the identity is $da = -\gamma a^2 dt + b dB_t$. Hence

$$\frac{d}{dt} \mathbb{E}a(t, X_t)^2 = \mathbb{E}b(t, X_t)^2 - 2\gamma(t)\mathbb{E}a(t, X_t)^3 \quad \text{for almost every } t < 1.$$

The left side is zero by Theorem 1.1. Convexity gives $a \geq 0$, and the same theorem gives $\mathbb{E}a^2 = 1$, so $\mathbb{E}a^3 \geq (\mathbb{E}a^2)^{3/2} = 1$. The ratio on the right of (6.8) is therefore well defined and continuous on every compact time strip. Its almost-everywhere equality with γ extends to every time by right continuity. In particular γ is continuous. At time zero, evenness gives $b(0, 0) = 0$, so the formula yields $\gamma(0) = 0$.

To bootstrap regularity without assuming time derivatives of γ , consider all polynomial moments of the spatial derivatives at once. For a polynomial $P = P(y_1, \dots, y_m)$, define

$$M_P(t) = \mathbb{E}P(f_1(t, X_t), \dots, f_m(t, X_t)).$$

In formal variables $y_1, y_2, \dots$, set

$$\begin{aligned} D_k(y) &= -\frac{1}{2} \sum_{j=1}^{k-1} \binom{k}{j} y_{j+1} y_{k-j+1}, \\ Q_P(y) &= \frac{1}{2} \sum_{k,l=1}^m (\partial_k \partial_l P)(y) y_{k+1} y_{l+1}, \\ R_P(y) &= \sum_{k=1}^m (\partial_k P)(y) D_k(y). \end{aligned}$$

These are polynomials in finitely many variables, with $D_1 = 0$. Itô's product rule and (6.10) give

$$M_P(t) - M_P(0) = \int_0^t (M_{Q_P}(s) + \gamma(s)M_{R_P}(s)) \, ds. \quad (6.11)$$

Every such moment is finite and continuous on each compact strip. Since γ is continuous, (6.11) makes every M_P continuously differentiable; the ratio (6.8) then makes γ continuously differentiable. More generally, if γ and all these moments are C^k , the integrand in (6.11) is C^k for every P , making all moments C^{k+1} . The same ratio makes γ C^{k+1} . Induction proves smoothness, including right derivatives at zero.

Smoothness and $\gamma(0) = 0$ identify its Stieltjes measure with $\gamma'(t) \, dt$, and monotonicity gives $\gamma' \geq 0$. It remains to prove strict positivity away from zero. Set $d = f_4$. The polynomial-moment identities give

$$\frac{d}{dt} \mathbb{E}b^2 = \mathbb{E}d^2 - 6\gamma \mathbb{E}(ab^2), \quad \frac{d}{dt} \mathbb{E}a^3 = 3\mathbb{E}(ab^2) - 3\gamma \mathbb{E}a^4.$$

Differentiating $\mathbb{E}b^2 = 2\gamma \mathbb{E}a^3$ therefore yields

$$2\gamma' \mathbb{E}a^3 = \mathbb{E}d^2 - 12\gamma \mathbb{E}(ab^2) + 6\gamma^2 \mathbb{E}a^4. \quad (6.12)$$

All factors in this identity are evaluated at (t, X_t) .

Fix $t \in (0, 1)$ and put $c = \gamma(t)$. Full support implies $c > 0$. We next apply the integrated coercivity estimate at this fixed time, although γ need not be constant on a neighborhood of t . Choose $\delta_n \downarrow 0$ with $[t - \delta_n, t + \delta_n] \subset (0, 1)$. Replace γ on $[t - \delta_n, t + \delta_n]$ by the value c . This preserves monotonicity, since $\gamma(s) \leq c$ for $s < t$ and $\gamma(s) \geq c$ for $s > t$, and the change tends to zero in L^1 . Outside that interval, truncate at a height greater than c , impose a positive lower cutoff less than c , and approximate separately on the left and right by nondecreasing finite step functions with values at most c and at least c , respectively. Let the truncation height tend to infinity, the lower cutoff tend to zero, and the discretization error tend to zero. The resulting positive bounded nondecreasing coefficients γ_n converge to γ in L^1 and equal c on a neighborhood of t for every n .

Use terminal data $M_n^{-1} \log(2 \cosh(M_n x))$, with $M_n \geq \|\gamma_n\|_\infty$ and $M_n \rightarrow \infty$, and let a_n, b_n, d_n and X^n denote the corresponding spatial derivatives and diffusion. At time t , the scaled derivatives in Proposition 4.2 are $v_n = ca_n$, $w_n = cb_n$, and $z_n = cd_n$. Thus that proposition and Lemmas 3.2–3.4 give

$$\mathbb{E}[d_n^2 - 12ca_nb_n^2 + 6c^2a_n^4](t, X_t^n) \geq \frac{c^2}{1000} \mathbb{E}a_n(t, X_t^n)^4.$$

The derivative bounds and convergence on a fixed compact strip from Lemmas 2.1 and 2.2 do not depend on the widths δ_n of the artificial plateaus. They therefore allow passage to the limit in every expectation in the last display. Combine the limiting inequality with (6.12) and $c = \gamma(t)$ to obtain (6.9). Its right side is strictly positive because $c > 0$ and $\mathbb{E}a^4 \geq (\mathbb{E}a^2)^2 = 1$. The proof asserts only $\gamma'(0) \geq 0$ at the left endpoint. $\square$

7 Martingale representation and optimization consequences

Full support gives the gradient martingale exactly the second-moment profile of Brownian motion: $\mathbb{E}u_\gamma(t, X_t)^2 = t$. Its correlation with the driving Brownian motion recovers the ground-state value. We first prove this identity and then approximate it using a single $\{-1, 1\}$ -valued function of finitely many independent Gaussians. The gradient-martingale and normalized Euler approach is developed in Montanari [14, Sections 2–3] and El Alaoui, Montanari, and Sellke [9, Sections 5–6 of the cited preprint]. Here Theorem 1.1 supplies the required zero-temperature second-moment normalization.

Corollary 7.1 (Martingale representation of the value). *Let γ be a minimizer, let X be its diffusion, and write $u = \Phi_{\gamma,x}$ and $a = \Phi_{\gamma,xx}$. There is a bounded martingale terminal value U_1 such that $u(t, X_t) \rightarrow U_1$ almost surely and in L^2 as $t \uparrow 1$, with $U_1^2 = 1$ almost surely. Moreover,*

$$\begin{aligned} P_* &= \mathbb{E}|X_1| - \int_0^1 t\gamma(t) dt \\ &= \mathbb{E}[U_1 B_1] = \int_0^1 \mathbb{E}a(t, X_t) dt. \end{aligned} \tag{7.1}$$

The last integral is finite.

Proof. Lemma 2.2 gives the bounded martingale $u(t, X_t) = \int_0^t a(s, X_s) dB_s$. By Theorem 1.1, its second moment is t and $\mathbb{E}a(t, X_t)^2 = 1$. Martingale convergence therefore gives U_1 with $|U_1| \leq 1$ and $\mathbb{E}U_1^2 = 1$, hence $U_1^2 = 1$ almost surely. The diffusion converges to X_1 almost surely and in L^2 , because its drift has deterministic integrable bound γ .

The uniform terminal bound (2.1) and convexity show that $u(t, x_t) \rightarrow \text{sgn}(x)$ whenever $x_t \rightarrow x \neq 0$. Indeed, for a fixed small $h > 0$, the derivative of a convex function at x_t lies between its left and right difference quotients of step h ; uniform convergence to $|\cdot|$ makes both limits equal to the sign when the interval stays away from zero. Consequently $U_1 X_1 = |X_1|$ almost surely, with no need to exclude $X_1 = 0$.

Itô's formula along the optimal diffusion, justified on compact strips as in Lemma 2.2, gives

$$\mathbb{E}\Phi_\gamma(t, X_t) = \Phi_\gamma(0, 0) + \frac{1}{2} \int_0^t \gamma(s) \mathbb{E}u(s, X_s)^2 ds.$$

Let $t \uparrow 1$. The terminal bound and L^1 convergence of X_t identify the left side with $\mathbb{E}|X_1|$. Substitute $\mathbb{E}u(s, X_s)^2 = s$ and the definition of $\mathcal{P}$ to obtain the first line of (7.1). Since $X_1 = B_1 + \int_0^1 \gamma(s)u(s, X_s) ds$, martingality and Fubini's theorem yield

$$\mathbb{E}[U_1 X_1] = \mathbb{E}[U_1 B_1] + \int_0^1 \gamma(s) \mathbb{E}[u(s, X_s)^2] ds.$$

All terms are integrable because $|u|, |U_1| \leq 1$ and $\gamma \in L^1$. Finally the L^2 stochastic-integral identity at time one implies $\mathbb{E}[U_1 B_1] = \int_0^1 \mathbb{E}a(s, X_s) ds$. The integral is absolutely bounded by one using $\mathbb{E}|a(s, X_s)| \leq (\mathbb{E}a(s, X_s)^2)^{1/2} = 1$. $\square$

The last expression in (7.1) is the expected covariation of the gradient martingale with its driving Brownian motion. To approximate it, replace the diffusion by an Euler scheme on a fixed strip $[0, T]$, where $T < 1$. The continuum identity $\mathbb{E}a(t, X_t)^2 = 1$ makes each martingale increment have variance equal to its time step. The Euler scheme has only an approximate version of this normalization, so we normalize its Gaussian-weighted directions to unit variance.

One additional independent Gaussian then rounds the Euler terminal gradient to a spin while preserving its conditional mean. The coefficients A_j below measure this spin's correlations with the normalized martingale directions; the coefficients B_i measure its correlations with the original Gaussian coordinates. Their adjacent-index product converges to the same covariation integral. These are scalar Gaussian coefficients, not an implementation on the finite-spin disorder.

On this fixed strip, u, a and all their spatial derivatives are bounded; the Gaussian increments remain unbounded. We first refine the mesh and only afterwards let T approach one. The mesh count N is distinct from the spin-system size n in the introduction.

Proposition 7.2 (Finite Gaussian coefficients). *Fix $0 < T < 1$, a minimizer γ , and u, a as above. For an integer $N \geq 1$ put $\Delta = T/N$ and $t_j = j\Delta$. On independent standard normal variables $Z_1, \dots, Z_{N+1}$, define*

$$Y_0 = 0, \quad Y_j = Y_{j-1} + \sqrt{\Delta}Z_j + \Delta\gamma(t_{j-1})u(t_{j-1}, Y_{j-1}), \quad 1 \leq j \leq N, \quad (7.2)$$

$$c_j = (\mathbb{E}a(t_{j-1}, Y_{j-1})^2)^{-1/2} > 0, \quad g_j = c_j a(t_{j-1}, Y_{j-1})Z_j. \quad (7.3)$$

For all sufficiently large N these constants are finite and positive. For these mesh counts, let F_G denote the standard normal distribution function and set

$$F_N = \text{sgn}(u(T, Y_N) + 2F_G(Z_{N+1}) - 1), \\ A_j = \mathbb{E}[F_N g_j], \quad B_i = \mathbb{E}[F_N Z_i], \quad A_0 = 0.$$

Here $1 \leq j \leq N$ and $1 \leq i \leq N+1$ in the definitions of A_j and B_i . The value of the sign at zero is immaterial. Then

$$\lim_{N \rightarrow \infty} \sum_{j=0}^N A_j B_{j+1} = \int_0^T \mathbb{E}a(t, X_t) dt. \quad (7.4)$$

Letting $T \uparrow 1$ on the right gives P_ .*

Proof. Euler convergence and normalization. Use Brownian increments $\sqrt{\Delta}Z_j = B_{t_j} - B_{t_{j-1}}$ to couple (7.2) to X up to time T . Lemma 2.1, on a slightly larger strip, bounds all spatial derivatives of u, a . Their almost-everywhere time derivatives given by the PDE are bounded there, so u, a are uniformly Lipschitz in time and space. The drift γu is spatially Lipschitz with a common bound. For its time dependence, it is enough to use that γ is bounded and monotone. If $\pi_N(s) = t_{j-1}$ on $[t_{j-1}, t_j)$, then

$$\int_0^T |\gamma(s) - \gamma(\pi_N(s))| ds \leq \Delta(\gamma(T) - \gamma(0)).$$

Also $\mathbb{E}|X_s - X_{\pi_N(s)}|^2 \leq C\Delta$. Subtract the two integral drift equations at the grid points, use these bounds and the Lipschitz bounds, and apply discrete Gronwall. This gives

$$\mathbb{E} \max_{j \leq N} |Y_j - X_{t_j}|^2 \rightarrow 0. \quad (7.5)$$

More explicitly, the L^2 norm of the sum of local drift errors is at most $C\sqrt{\Delta} + C\Delta$; the remaining errors satisfy $e_j \leq C\Delta \sum_{i < j} e_i + o(1)$, which proves the claim.

Uniform convergence in (7.5), boundedness and Lipschitz continuity of a , and $\mathbb{E}a(t, X_t)^2 = 1$ imply $\max_{j \leq N} |c_j - 1| \rightarrow 0$. The g_j form an orthonormal family in L^2 : they have unit second moment by definition and zero conditional mean given the previous increments. Itô isometry, (7.5), and the same continuity estimates give

$$M_N := \sum_{j=1}^N \sqrt{\Delta} a(t_{j-1}, Y_{j-1})Z_j = u(T, Y_N) + o_{L^2}(1).$$

Rounding and the two coefficient vectors. The independent random variable $F_G(Z_{N+1})$ is uniform on $(0, 1)$. For any $v \in [-1, 1]$, the sign of $v + 2U - 1$, with U uniform on $(0, 1)$, has mean v . Hence $\mathbb{E}[F_N | Z_1, \dots, Z_N] = u(T, Y_N)$ exactly. This preserves every correlation with a function of the first N Gaussians. Bessel's inequality applied to $u(T, Y_N) - M_N$ against the two orthonormal families $(g_j)_{j=1}^N$ and $(Z_i)_{i=1}^N$ yields

$$(A_j)_{j=1}^N = (\sqrt{\Delta}/c_j)_{j=1}^N + o_{\ell^2}(1), \\ (B_i)_{i=1}^N = (\sqrt{\Delta} \mathbb{E}a(t_{i-1}, Y_{i-1}))_{i=1}^N + o_{\ell^2}(1).$$

For the second equality, terms of M_N before i are independent of Z_i , and those after i have zero conditional mean; the i th term contributes the stated expectation. The first equality follows from orthogonality of the g_j and $M_N = \sum_j \sqrt{\Delta} g_j / c_j$.

The shifted product. The first coefficient vector is asymptotically the constant vector with entries $\sqrt{\Delta}$, because $c_j \rightarrow 1$ uniformly. The second has entries $\sqrt{\Delta}$ times the expected Hessian at the preceding mesh point. Pairing A_j with B_{j+1} therefore selects the expected Hessian at t_j , which is a Riemann-sum value for the target integral. To justify the accumulated errors, the vectors in both displays have bounded ℓ^2 norm. Cauchy–Schwarz shows that their errors contribute $o(1)$ to the shifted scalar product with $1 \leq j < N$. The first display also gives $A_N = \sqrt{\Delta}/c_N + o(1) = o(1)$. The remaining term therefore obeys $|A_N B_{N+1}| \leq |A_N| = o(1)$, since $|B_{N+1}| \leq 1$. Thus the sum in (7.4) differs by $o(1)$ from

$$\Delta \sum_{j=1}^{N-1} \frac{\mathbb{E}a(t_j, Y_j)}{c_j}.$$

This converges to the stated integral by (7.5), uniform convergence of c_j to one, and continuity of $t \mapsto \mathbb{E}a(t, X_t)$. Corollary 7.1 gives its limit as $T \uparrow 1$. $\square$

7.1 An application of the existing optimization theorem

The full-support theorem also verifies the hypothesis of an existing algorithmic result. We use the operation count of El Alaoui, Montanari, and Sellke [9, Remark 2.1, Theorem 3, and Corollary 2.2 of the cited preprint]: real sums and products have unit cost, and the scalar functions and normalization constants are those prescribed in their algorithm for the chosen accuracy. This is an instance-size bound; the argument below supplies no separate complexity estimate for numerical preprocessing of those functions or for finite-precision arithmetic.

Corollary 7.3 (Asymptotically optimal SK optimization). *For every fixed $\epsilon > 0$, there is a randomized algorithm with operation count at most $C(\epsilon)n^2$ in the model just described, where $C(\epsilon)$ is independent of n , that takes the Gaussian couplings of H_n as input and returns $\sigma^{\text{alg}} \in \{-1, 1\}^n$ such that*

$$\Pr \left\{ H_n(\sigma^{\text{alg}}) \geq \max_{\sigma \in \{-1, 1\}^n} H_n(\sigma) - \epsilon n \right\} \rightarrow 1 \quad (n \rightarrow \infty).$$

The probability includes the couplings and the algorithm’s randomness.

Proof. Theorem 1.1 gives a strictly increasing minimizer in $\mathcal{U}$, exactly the no-overlap-gap hypothesis in [9, Assumption 2] for $\xi(t) = t^2/2$. We apply their Corollary 2.2, with its operation count from Theorem 3.

To match their Gaussian quadratic input, form a symmetric matrix A with $A_{ij} = J_{ij}/\sqrt{n}$ for $i < j$ and independent diagonal entries $A_{ii} = \sqrt{2/n} G_i$, where the G_i are independent standard Gaussians. The quadratic extension $\hat{H}_n(x) = \frac{1}{2} x^\top A x$ has covariance $(2n)^{-1}(\sigma^\top \tau)^2$ on the hypercube, and

$$\hat{H}_n(\sigma) = H_n(\sigma) + \frac{1}{\sqrt{2n}} \sum_{i=1}^n G_i.$$

The second term is independent of σ , so every optimization gap is identical for H_n and $\hat{H}_n$.

Evaluating $\nabla \hat{H}_n(x) = Ax$ costs $O(n^2)$ operations. The part with distinct indices is $H_n(x)$, and evaluating one coordinate of its gradient costs $O(n)$. Thus the bound $C(\epsilon)(\chi + n) + n\chi_1$ in the cited theorem is $C(\epsilon)n^2$, after changing the constant. Its probability guarantee and the equality of the optimization gaps give the assertion. $\square$

This corollary invokes the finite-disorder message-passing theorem of El Alaoui, Montanari, and Sellke. Proposition 7.2 supplies the separate scalar approximation proved above.

References

- [1] Antonio Auffinger and Wei-Kuo Chen. On properties of Parisi measures. *Probability Theory and Related Fields*, 161(3–4):817–850, 2015. Preprint version cited: <https://arxiv.org/abs/1303.3573v1>.
- [2] Antonio Auffinger and Wei-Kuo Chen. The Parisi formula has a unique minimizer. *Communications in Mathematical Physics*, 335(3):1429–1444, 2015. Preprint version cited: <https://arxiv.org/abs/1402.5132v3>.
- [3] Antonio Auffinger and Wei-Kuo Chen. Parisi formula for the ground state energy in the mixed p -spin model. *The Annals of Probability*, 45(6B):4617–4631, 2017. Preprint version cited: <https://arxiv.org/abs/1606.05335v2>.
- [4] Antonio Auffinger, Wei-Kuo Chen, and Qiang Zeng. The SK model is infinite step replica symmetry breaking at zero temperature. *Communications on Pure and Applied Mathematics*, 73(5):921–943, 2020. Preprint version cited: <https://arxiv.org/abs/1703.06872v2>.
- [5] Michelle Boué and Paul Dupuis. A variational representation for certain functionals of Brownian motion. *The Annals of Probability*, 26(4):1641–1659, 1998.
- [6] Anton Bovier and Anton Klimovsky. The Aizenman–Sims–Starr and Guerra’s schemes for the SK model with multidimensional spins. *Electronic Journal of Probability*, 14(8):161–241, 2009. Preprint version cited: <https://arxiv.org/abs/0802.3467v2>.
- [7] Hong-Bin Chen. FRSB in the SK spin glass: convergence to full-interval support at zero temperature, 2026. <https://arxiv.org/abs/2607.18032v1>.
- [8] Wei-Kuo Chen, Madeline Handschy, and Gilad Lerman. On the energy landscape of the mixed even p -spin model. *Probability Theory and Related Fields*, 171:53–95, 2018. Preprint version cited: <https://arxiv.org/abs/1609.04368v4>.
- [9] Ahmed El Alaoui, Andrea Montanari, and Mark Sellke. Optimization of mean-field spin glasses. *The Annals of Probability*, 49(6):2922–2960, 2021. Preprint version cited: <https://arxiv.org/abs/2001.00904v1>.
- [10] Francesco Guerra. Broken replica symmetry bounds in the mean field spin glass model. *Communications in Mathematical Physics*, 233(1):1–12, 2003.
- [11] Aukosh Jagannath and Ian Tobasco. A dynamic programming approach to the Parisi functional. *Proceedings of the American Mathematical Society*, 144(7):3135–3150, 2016. Preprint version cited: <https://arxiv.org/abs/1502.04398v4>.
- [12] Aukosh Jagannath and Ian Tobasco. Some properties of the phase diagram for mixed p -spin glasses. *Probability Theory and Related Fields*, 167:615–672, 2017. Preprint version cited: <https://arxiv.org/abs/1504.02731v2>.

- [13] Patrick Lopatto. Full replica symmetry breaking in the Sherrington–Kirkpatrick model, 2026. Version 3; displayed manuscript date August 24, 2026. <https://arxiv.org/abs/2607.11756v3>.
- [14] Andrea Montanari. Optimization of the Sherrington–Kirkpatrick hamiltonian, 2019. Version 2, April 5, 2019, <https://arxiv.org/abs/1812.10897v2>. Published in SIAM Journal on Computing 54(4), FOCS19-1–FOCS19-38 (2025); first online publication January 7, 2021.
- [15] Giorgio Parisi. Infinite number of order parameters for spin-glasses. *Physical Review Letters*, 43(23):1754–1756, 1979.
- [16] Giorgio Parisi. Order parameter for spin-glasses. *Physical Review Letters*, 50(24):1946–1948, 1983.
- [17] David Sherrington and Scott Kirkpatrick. Solvable model of a spin-glass. *Physical Review Letters*, 35(26):1792–1796, 1975.
- [18] Michel Talagrand. The Parisi formula. *Annals of Mathematics*, 163(1):221–263, 2006.
- [19] Yuxin Zhou. Existence of full replica symmetry breaking for the Sherrington–Kirkpatrick model at low temperature. *Communications on Pure and Applied Mathematics*, 79(7):1746–1770, 2026.